

Naive vs. bias-corrected g-estimation

Three-time simulation, variation-independent contrasts that do not form an SNMM (Figure 4)

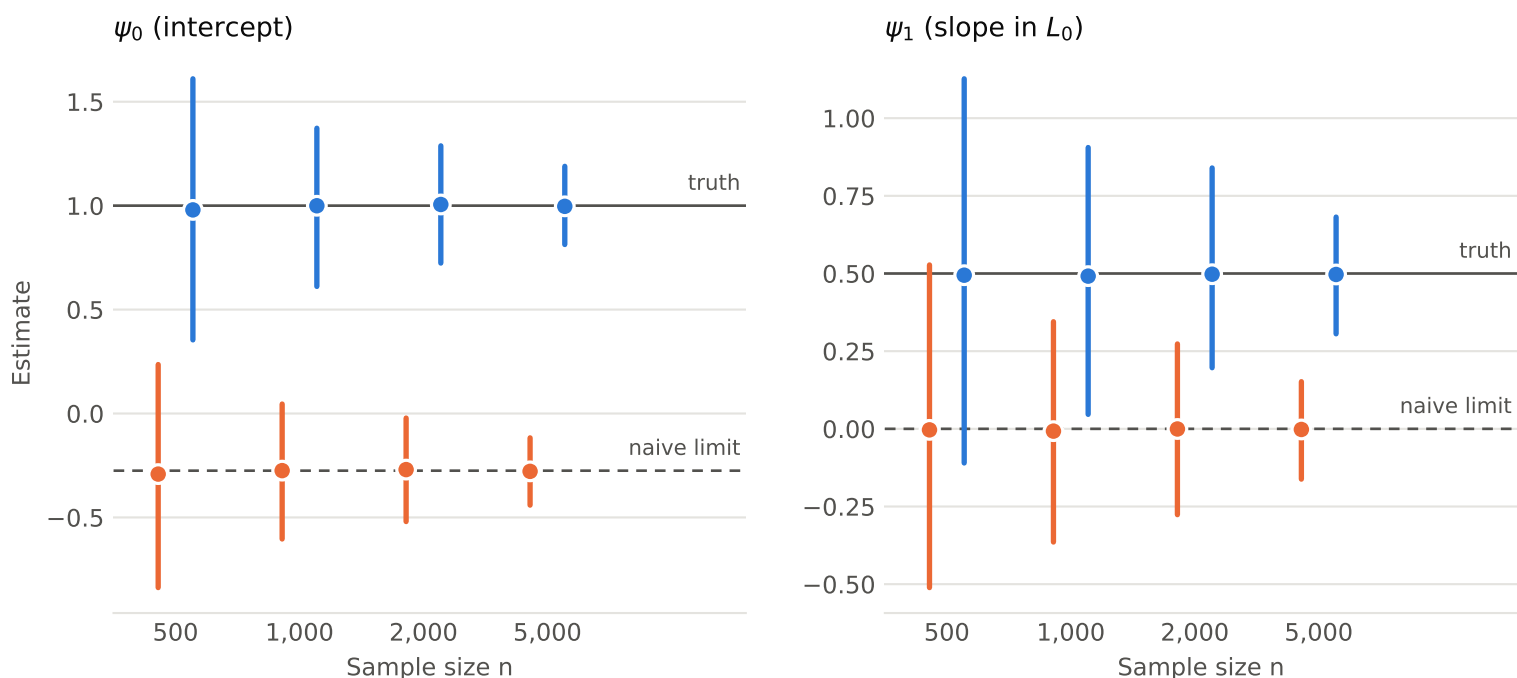
Target: the Trial 0 blip $\gamma_0(L_0) = E(Y^{1,1,1} - Y^{0,1,1} | L_0) = \psi_0 + \psi_1 L_0$, with truth (1, 0.5).

Naive g-estimation (Equation 4) blips down with the Trial 1 regime, so it is consistent for $E(Y^{1,1,0} - Y^{0,1,0} | L_0) = -0.275 + 0 L_0$ instead. The corrected estimator solves the Theorem 4 influence-function equation.

γ_1, γ_2 are treated as known; π_0, π_1, q, μ_0 are fitted with correctly specified parametric models.

1,000 replicates per sample size.

Naive estimates settle on the wrong contrast; corrected estimates on the truth



● Naive g-estimation ● Bias-corrected g-estimation Dots: Monte Carlo mean. Bars: central 95% of the estimates.

Estimated nuisance functions

n	Method	ψ_0 bias	ψ_0 SD	ψ_0 SE	ψ_0 cover	ψ_1 bias	ψ_1 SD	ψ_1 SE	ψ_1 cover
500	Naive	-1.291	0.271	0.255	0.2%	-0.503	0.275	0.270	52.2%
500	Bias-corrected	-0.020	0.316	0.297	92.3%	-0.005	0.318	0.314	95.2%
1,000	Naive	-1.275	0.172	0.180	0.0%	-0.507	0.188	0.190	24.1%
1,000	Bias-corrected	-0.001	0.201	0.210	96.5%	-0.008	0.219	0.221	95.2%
2,000	Naive	-1.270	0.130	0.128	0.0%	-0.500	0.138	0.135	4.9%
2,000	Bias-corrected	0.006	0.149	0.149	95.7%	-0.002	0.160	0.157	94.3%
5,000	Naive	-1.278	0.081	0.081	0.0%	-0.502	0.083	0.085	0.0%
5,000	Bias-corrected	-0.003	0.094	0.094	94.3%	-0.003	0.097	0.100	96.3%

SD: empirical standard deviation. SE: mean sandwich standard error. Cover: coverage of the truth by 95% Wald intervals.

Oracle nuisance functions and design

Same replicates, with the true π_0, π_1, q, μ_0 plugged in

Oracle nuisance functions

n	Method	ψ_0 bias	ψ_0 SD	ψ_0 SE	ψ_0 cover	ψ_1 bias	ψ_1 SD	ψ_1 SE	ψ_1 cover
500	Naive	-1.291	0.270	0.256	0.4%	-0.503	0.274	0.271	52.5%
500	Bias-corrected	-0.020	0.315	0.298	92.4%	-0.005	0.317	0.316	96.0%
1,000	Naive	-1.275	0.172	0.181	0.0%	-0.507	0.188	0.191	25.0%
1,000	Bias-corrected	-0.001	0.202	0.211	97.1%	-0.008	0.220	0.222	95.2%
2,000	Naive	-1.270	0.130	0.128	0.0%	-0.501	0.137	0.135	4.7%
2,000	Bias-corrected	0.006	0.149	0.149	95.7%	-0.003	0.160	0.157	94.3%
5,000	Naive	-1.278	0.081	0.081	0.0%	-0.502	0.083	0.085	0.0%
5,000	Bias-corrected	-0.003	0.094	0.094	94.4%	-0.003	0.097	0.100	96.0%

Data generating mechanism

$L_0 \sim N(0, 1), A_0 \sim \text{Bern}\{\text{expit}(0.5L_0)\}$

$L_1 = 0.5L_0 + 0.5A_0 + N(0, 1), A_1 \sim \text{Bern}\{\text{expit}(0.5L_1 - 0.3A_0)\}$

$L_2 = 0.5L_1 + 0.5A_1 + 0.3A_0 + N(0, 1), A_2 \sim \text{Bern}\{\text{expit}(0.5L_2 - 0.3A_1)\}$

$Y = L_0 + L_1 + L_2 + A_2\gamma_2(\bar{S}_2) + A_1d_1(\bar{S}_1) + A_0d_0(S_0) + N(0, 1)$

d_1, d_0 are back-solved so the Trial 1 and Trial 0 contrasts equal the blips below.

Causal contrasts

Trial 2: $E(Y^{A_0, A_1, 1} - Y^{A_0, A_1, 0} | \bar{S}_2) = \gamma_2 = 1 + 0.5L_2 + A_0(1 + 0.5L_0)$

Trial 1: $E(Y^{A_0, 1, 0} - Y^{A_0, 0, 1} | \bar{S}_1) = \gamma_1 = 1 + 0.5L_1, h_2^1 = 1 - A_1$

Trial 0: $E(Y^{1, 1, 1} - Y^{0, 1, 1} | S_0) = \gamma_0 = 1 + 0.5L_0, h_1^0 = h_2^0 = 1$

Estimating equations

Both solve $P_n[c(L_0)\{A_0 - \pi_0(L_0)\}\{H - A_0\gamma_0(L_0; \psi) - \mu_0(L_0)\}] = 0$ with $c(L_0) = (1, L_0)^\top$.

Naive: $H = Y - (A_2 - h_2^1)\gamma_2 - (A_1 - h_1^0)\gamma_1$

Corrected: $H = \text{naive} + R(\Delta\gamma_2 - q) + q, R = A_1/\pi_1(\bar{S}_1), \Delta = h_2^0 - h_2^1, q = E(\Delta\gamma_2 | \bar{S}_1, A_1 = 1)$

Naive limit

$\gamma_0(L_0) - \{\theta_{21}\tau + \theta_{22} + \theta_{23}L_0\}$, where $\tau = 0.55$ is the effect of A_0 on $E(L_2)$ with A_1 fixed and $\theta_{21}, \theta_{22}, \theta_{23} = 0.5, 1, 0.5$ are the L_2, A_0 and A_0L_0 coefficients of γ_2 . The bias is how much A_0 modifies the effect of A_2 .

1,000 replicates per sample size. Standard errors treat the nuisance functions as fixed.